P1 audit of UrbanSound8K, two-page summary

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The full notebook, `IDIG4120_P1_Kivle_Stubø.ipynb`, stays as the extended record with every table, plot and audio player. Every number below comes from its run over the full release.

The dataset

UrbanSound8K holds 8,732 WAV excerpts of urban sound, at most 4 seconds each, in 10 classes (air_conditioner through street_music) and 10 predefined folds for cross-validation, plus one metadata CSV. We took it from Kaggle on 2026-08-25 under CC BY-NC 4.0. Recording quality varies across the collection, so the audit checks every file header against the expected bit depth, channel count and sample rate before it counts anything. Nothing about this kind of audio raises a security concern.

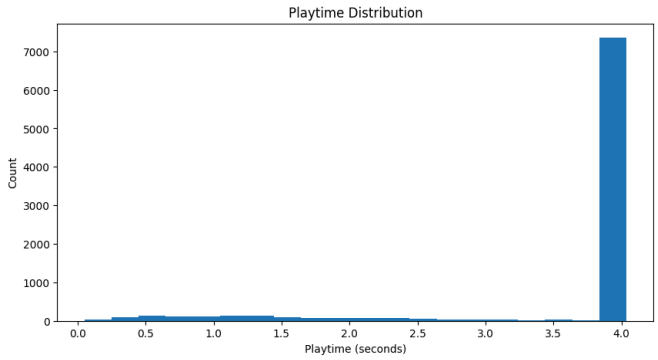
What the audit found

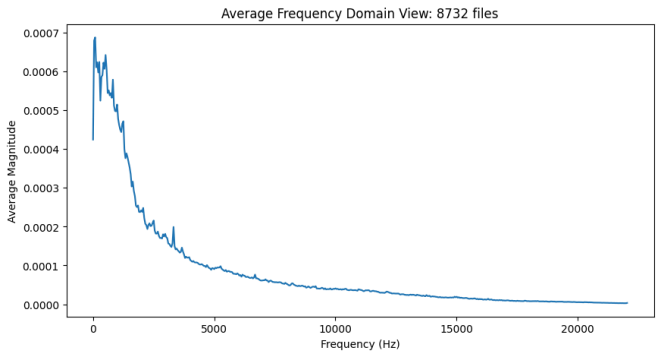
All 8,732 files load and decode, zero failures, and the CSV matches every one of them. The encodings vary:

Subtype	Files	Decodes to
PCM_16	5,758	int16
PCM_24	2,753	int32
FLOAT	169	float32
PCM_U8	43	uint8
ADPCM (IMA and MS)	9	int16

Samples come out as arrays shaped `(frames, channels)`: 739 files mono, 7,993 stereo, averaging 173,864 frames. Amplitudes are normalized to full scale, -1 to 1: integer PCM is mapped by its own range, and floating-point samples pass through unchanged. On disk the release weighs 6,760.26 MiB, about 812 kB a file. Sample rates vary from file to file across 11 rates, 8,000 to 192,000 Hz with the mean near 48,457 Hz, and nothing in this project ever resamples. Clips run up to 4 seconds, 3.61 on average, 8.75 hours in total.

The time view and the frequency view, both with labelled axes:





Two experiments

Quantisation to int16. Our prediction: files at or below 16 bits should come back bit-identical, their extra precision is false, and only the 24-bit and float files should really change. That is what happens. 5,810 of 8,732 files (PCM_16, PCM_U8 and the ADPCM files) return RMSE exactly 0. The 2,922 wider files move barely at all: mean RMSE 8.7e-6 for FLOAT and 8.2e-6 for PCM_24, the worst single file at 1.3e-5. Converting everything to int16 cuts storage from 6,760.26 to 5,581.96 MiB, a 17.43% saving, and it fixes the baseline P2 measures against: P2 compresses int16 samples, so its ratios are measured against the 16-bit PCM payload, not the on-disk byte size.

Fourier pruning. Our prediction: removing every coefficient below a fraction of each channel's strongest component should shrink the sparse representation, and the reconstruction error should grow slowly at first. The sweep over all 873 fold1 files (769.76 MiB) confirms it:

Threshold	Coefficients kept	Sparse size	Reduction	Mean RMSE
0%	100%	1,226.69 MiB	-59.4%	0.000000
0.1%	42.3%	530.32 MiB	31.1%	0.001115
0.5%	22.2%	281.05 MiB	63.5%	0.004980

Threshold	Coefficients kept	Sparse size	Reduction	Mean RMSE
1%	15.6%	199.13 MiB	74.1%	0.008920
2%	10.0%	128.90 MiB	83.3%	0.015431
5%	4.4%	57.44 MiB	92.5%	0.028621

One artefact is worth naming: at threshold 0 the sparse file is 59% *larger* than the WAV, because storing indices and complex coefficients costs more than the signal itself until pruning actually removes something. Listened to at a 10% threshold, a clip keeping 0.85% of its coefficients is still clearly recognizable. The notebook plays the removed signal next to the reconstruction: what the error metric calls negligible, RMSE 0.005, is plainly audible content. A file can pass every structural check and still lose audible material, so the audit reports both checks.

Where the numbers come from

The loader (scipy as the primary decoder, soundfile as fallback, the CSV optional) validates and measures every file in one pass over the release. Rerunning it reproduces everything above.

Nikolai started P1 and Katharina reviewed the first part before we finished the collaborated sections together. Katharina ran the filtering experiment, Nikolai the quantisation. The generative-AI declaration sits next to this document as `IDIG4120-KI-Deklarasjon-P1.pdf`.